LearnHub

Lecture notes: by date and by subject (minor differences may occur between the two versions)

Class exercises and their solutions

Reading assignments: **the lowest grade will be excluded**

- Reading assignment 2: Chapters 3-4 (Assessments) due midnight next Monday (17/03/2025).
- You will have one week to work on each reading assignment every Monday.

Lab 4: today on CLT and CIs

16

Schedule

Sample case:

- To be discussed next Thursday in the lab session
- The related files will be shared over the weekend. You are supposed to carry out preliminary analyses and think about the questions
- We will first discuss the case; then, you will be given a multiple-choice quiz via LearnHub.

17

17

Schedule

Case exam: at **17:20-19:20** on March 27th in SNAB149 and SNAB220 (so those who fast can start iftar on time)

Groups will be assigned to one of these computer labs. You will be allowed to use only the computers in the labs (you cannot use your own computer)

Groups: You will work in groups of 6-7 students. Please form your groups by next Friday (March 21st) in Indr252Lab LearnHub

Information on case exams: Indr252Lab LearnHub

Sample case exams, reports: Indr252Lab LearnHub

Read all the information provided, and I will answer your questions during the lectures next week.

18

Today

Central limit theorem (Section 4.11)

Confidence Intervals (Chapter 5)

19

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$$

Central Limit Theorem (CLT)

$X_1, \ldots, X_n$ be independent identically distributed (iid) rvs with $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$. Then:

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \rightarrow Z_n \sim N(0,1) \quad \text{as} \quad n \rightarrow \infty,$$

$$P\left(\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \leq a\right) \xrightarrow{n\rightarrow\infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{a} e^{-\frac{x^2}{2}} dx$$

20

20

Central Limit Theorem

- Central Limit Theorem applet with different distributions.
- You will observe this in Python simulations in the lab exercises and assignments.
- Applet: https://onlinestatbook.com/stat_sim/sampling_dist/

21

21

$$\sigma^2 = \frac{(b-a)^2}{12}$$

Example for CLT

Suppose that a random variable X has a uniform distribution with parameters (10,70).

What is the mean and standard deviation of X? $\mu = 40,\ \sigma = 17.32$

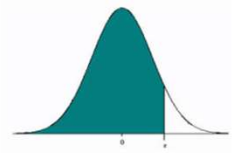
What is the approximate distribution of the sample mean of a random variable of size n=40?

Can we find its parameters?

What is the probability that this sample mean is larger than 44?

22

Table of Standard Normal Probabilities for Positive Z-scores



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

23

23

Example 23

If a fair coin is tossed 100 times, use the normal curve to approximate the probability that the number of heads is between 45 and 55 *inclusive*.

24

Normal approximation to Binomial
(DeMoivre-Laplace theorem)

We can have a slightly refined approximation by a small correction to account for the discreteness of the Binomial distribution.

P(*X*≤*x)* ≈ area under the normal curve to the left of *x* + 0.5

or equivalently:

$$P(k \le X \le l) \cong \Phi\left(\frac{l + 0.5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k + 0.5 - np}{\sqrt{np(1-p)}}\right)$$

25

25

Example 23

If a fair coin is tossed 100 times, use the normal curve to approximate the probability that the number of heads is between 45 and 55 *inclusive*.

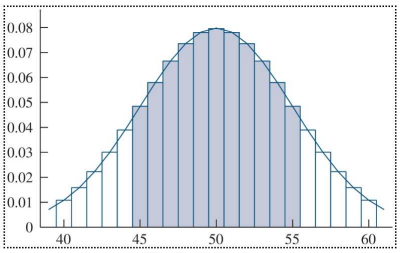


FIGURE 4.28 To compute $P(45 \leq X \leq 55)$, the areas of the rectangles corresponding to 45 and to 55 should be included. To approximate this probability with the normal curve, compute the area under the curve between 44.5 and 55.5.

26

Continuity Correction

The binomial distribution is discrete, while the normal distribution is continuous.

The continuity correction is an adjustment made when approximating a discrete distribution with a continuous one that can improve the accuracy of the approximation.

If you want to include the endpoints in your probability calculation, extend each endpoint by 0.5. Then, proceed with the calculation.

If you want to exclude the endpoints in your probability calculation, then include 0.5 less from each endpoint in the calculation.

27

Normal Approximation to the Poisson

If $X \sim \text{Poisson}(\lambda)$, where $\lambda > 10$, then $X \sim N(\lambda, \lambda^2)$.

Continuity correction for Poisson?

28